



OPEN Plant growth-promoting microorganisms and phosphorus levels influence growth, nutrient uptake, and secondary metabolites in hydroponic basil

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Plant growth promoting microorganisms (PGPMs) represent an environmentally sustainable alternative to chemical fertilizers, yet their application in soilless cultivation systems remains insufficiently studied. This study investigated the individual effects of four PGPMs, *Bacillus subtilis*, *Pantoea agglomerans*, *Serendipita indica*, *Funneliformis mosseae*, and three phosphorus concentrations (10, 20, and 40 mg L⁻¹) on growth, nutrient uptake, and essential oils (EOs) production in hydroponically grown basil (*Ocimum basilicum* L.). Results indicated that reduced phosphorus availability restricted biomass production, whereas PGPM inoculation significantly enhanced plant growth and improved shoot and root P and K accumulation ($P \leq 0.01$). EOs content was strongly influenced by both microbial inoculation and P level: *S. indica* and *P. agglomerans* at 40 mg L⁻¹ P increased EOs percentage four- and three-fold, respectively, compared with controls. Chemical profiling revealed notable shifts in EOs composition, with estragole as the predominant component. Inoculation with *F. mosseae* at 40 mg L⁻¹ P yielded the highest estragole concentration (43.7%), while *B. subtilis* under the same condition maximized δ -cadinene, β -farnesene, α -bergamotene, and γ -cadinene levels. These findings highlight the potential of PGPMs to improve phosphorus use efficiency and enhance both yield and phytochemical quality in hydroponic basil, providing a practical and sustainable approach for reducing fertilizer inputs in soilless cultivation systems.

Keywords Arbuscular mycorrhizal fungi (AMF), Basil (*Ocimum basilicum* L.), Hydroponics, Plant growth promoting bacteria (PGPB), Phosphorus uptake, Secondary metabolites

Global population growth, climate change, and the reduction of arable land pose major challenges to food security. Soilless cultivation systems, particularly hydroponics, have emerged as sustainable alternatives to conventional soil-based agriculture, offering more efficient nutrient and water use. In recent years, increasing attention has been directed toward the incorporation of beneficial microorganisms into soilless systems as a strategy to compensate for the absence of native soil microbiota and enhance plant performance while reducing dependency on chemical fertilizers^{1,2}. Plant growth-promoting microorganisms (PGPM) are defined as a diverse group of bacteria and fungi capable of colonizing the rhizosphere or root tissues and enhancing plant growth and productivity through direct and indirect mechanisms³. Traditionally studied in soil-based systems, PGPMs include plant growth-promoting bacteria (PGPB), fungi such as arbuscular mycorrhizal fungi (AMF), and other root-associated microbes that improve nutrient acquisition, modulate phytohormone levels, suppress pathogens, and enhance tolerance to abiotic stresses^{3,4}. In soilless or hydroponic cultivation, although the growth medium lacks a soil matrix, the functional concept of PGPMs remains applicable when these microorganisms colonize the root surface and the nutrient solution interface, forming a rhizosphere-like environment. In such systems, PGPMs have been shown to solubilize nutrients, produce growth-stimulating compounds, and influence plant metabolic responses, thereby promoting plant growth and health much as they do in soil⁴. Recent studies highlight that PGPMs can improve nutrient uptake, mitigate abiotic stress, and enhance plant physiological

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performance under hydroponic conditions^{5–8}. Consequently, the application of beneficial microorganisms in hydroponic systems is increasingly regarded as an environmentally friendly approach for sustainable crop production.

Numerous rhizospheric bacteria and fungi are known to stimulate plant growth, yield, nutrient uptake, stress tolerance, and secondary metabolite accumulation in medicinal plants. Among these, arbuscular mycorrhizal fungi (AMF) represent a major component of the soil microbial community and contribute substantially to plant growth enhancement, stress protection, and soil quality improvement, with most terrestrial plants forming symbiotic associations with AMF⁹. Similarly, *Serendipita indica* (formerly *Piriformospora indica*) is an endophytic fungus capable of robust root colonization and axenic culture, functioning as a biotrophic mutualist analogous to AMF¹⁰. In addition to beneficial fungi, several bacterial genera have been extensively investigated for their plant growth-promoting potential. For example, *Bacillus* species such as *B. subtilis* are widely used as biofertilizers and biocontrol agents due to their ability to produce phytohormones, form stress-tolerant spores, and colonize plant roots efficiently¹¹. Likewise, phosphate-solubilizing bacteria such as *Pantoea agglomerans* enhance plant-available phosphorus and thereby improve plant growth and productivity¹². PGPMs promote plant development through multiple mechanisms, including nitrogen fixation, phosphorus solubilization, phytohormone synthesis, ACC deaminase activity, siderophore production, and pathogen suppression. Among these mechanisms, enhancement of phosphorus availability and uptake is particularly critical, especially in soilless cultivation systems where nutrient supply is entirely dependent on nutrient solution composition and management¹³.

Phosphorus (P) is one of the three primary mineral nutrients applied in agriculture and plays a fundamental role in plant growth and metabolism. Phosphorus deficiency results in stunted growth, poor root development, and inhibited flowering. At the cellular level, phosphorus serves as a structural and regulatory component of phospholipids, enzymes, and phosphoproteins, and is involved in essential metabolic processes such as energy transfer, amino acid and nucleotide synthesis, and nitrogen and sulfur assimilation^{14,15}. Phosphorus is also indispensable for adenosine triphosphate (ATP) synthesis and normal photosynthetic function. Its deficiency disrupts the photosynthetic electron transport chain, limits ATP production, and reduces carbon assimilation, thereby influencing metabolic flux toward both primary and secondary metabolic pathways¹⁶. Reviews focusing on fertilization and secondary metabolism in medicinal and aromatic plants have shown that phosphorus fertilization enhances the accumulation of phenolic compounds and EOs, highlighting the importance of adequate phosphorus availability for secondary metabolite production¹⁷.

PGPMs significantly contribute to phosphorus acquisition by enhancing root development and facilitating phosphorus mobilization and redistribution between inorganic and organic pools¹⁸. Rhizospheric bacteria and fungi release organic acids that reduce pH and solubilize phosphate compounds or chelate mineral ions, thereby increasing phosphorus availability. Moreover, the secretion of enzymes such as phytases and phosphatases further enhances phosphorus uptake by plants¹³. Improved phosphorus absorption mediated by PGPMs not only supports primary plant growth but also enhances the biosynthesis of secondary metabolites, including EOs, in medicinal plants. Phosphorus is particularly critical for EO production because isoprenoid biosynthesis relies on high-energy intermediates such as acetyl coenzyme A, ATP, and nicotinamide adenine dinucleotide phosphate (NADPH). At the biochemical level, phosphorus participates directly in the methylerythritol phosphate (MEP) and mevalonate (MVA) pathways of terpenoid biosynthesis, which depend on high-energy phosphate bonds in intermediates such as isopentenyl diphosphate (IPP) and dimethylallyl diphosphate (DMAPP), as well as ATP- and NADPH-dependent enzymatic reactions. By increasing phosphorus availability through solubilization of inorganic and organic P, secretion of phosphatase enzymes, and stimulation of root development, PGPM inoculation together with optimized phosphorus supply can act synergistically to enhance metabolic flux toward terpenoid synthesis, ultimately improving both growth and EO accumulation in hydroponically grown aromatic plants^{19,20}.

Several studies have demonstrated that microbial inoculation can significantly enhance basil growth and essential oil yield. Roshanpour et al.²¹ reported that a combined inoculation of *Bacillus*, *Azotobacter*, and *Azospirillum* increased fresh and dry biomass as well as EOs yield in basil. Similarly, Dasgan et al.²² showed that the application of beneficial bacteria and mycorrhizae in floating cultivation systems increased basil yield by approximately 18.94% and 13.94%, respectively, compared with the control. Raei et al.²³ also demonstrated that inoculation with *Azospirillum* improved growth traits and drought tolerance in basil plants.

Basil (*Ocimum basilicum* L.), an annual herb belonging to the Lamiaceae family, is widely cultivated for its culinary and medicinal value and has extensive applications in the food, pharmaceutical, and cosmetic industries²⁴. Although the beneficial effects of PGPMs on basil growth and productivity have been well documented^{25–30}, most studies have focused on soil-based cultivation systems. In contrast, the application of PGPMs in sustainable soilless or hydroponic systems, particularly in relation to phosphorus nutrition and secondary metabolite production, remains insufficiently explored.

Therefore, the present research aimed to investigate the effect of *Bacillus subtilis*, *Pantoea agglomerans*, *Serendipita indica* (formerly *Piriformospora indica*), *Funneliformis mosseae* (formerly *Glomus mosseae*) and different phosphorus concentrations on the growth, physiological traits, and essential oil production of basil under hydroponic conditions. This work seeks to provide insights into the potential integration of PGPMs and optimized phosphorus management as sustainable strategies to improve basil productivity and quality in soilless cultivation systems.

Results

Plant growth and biomass production

The growth of basil plants was significantly affected by both phosphorus levels and PGPM inoculation ($P \leq 0.01$; Table 1). Reduced phosphorus concentration resulted in a marked decline in shoot and root fresh weight, dry

Means square								
Value sources	df	Shoot FW	Shoot DW	Leaf area	RWC	Root FW	Root DW	Root volume
Replication	2	16.91 ^{ns}	10.50 ^{ns}	30,380.35 ^{ns}	0.79 ^{ns}	9.35 ^{ns}	0.17 ^{ns}	12.92 ^{ns}
Microorganisms (a)	4	253.32**	20.35**	497,292.23**	352.69**	395.29**	3.91**	815.49**
Phosphorus (b)	2	15,767.08**	491.77**	6,062,305.79**	314.99**	544.84**	16.40**	1070.42**
a × b	8	77.69 ^{ns}	2.16 ^{ns}	187,035.05**	93.40**	16.61 ^{ns}	0.42 ^{ns}	54.44 ^{ns}
Error	28	37.78	3.25	49,286.54	12.44	15.21	0.42	33.15
Cv (%)		3.55	9.84	10.61	5.13	9.74	18.75	9.36

Table 1. Analysis of variance (ANOVA), for the effects of plant growth-promoting microorganisms (PGPMs) and different phosphorus concentrations on growth parameters of hydroponically grown basil plants. ^{ns}non-significant; *significant at $p < 0.05$; **significant at $p < 0.01$.

Treatment		Shoot FW(g)	Shoot DW(g)	Root FW(g)	Root DW(g)	Root volume (mL)
Microorganism	Control	165.66 ± 35.55 c	16.14 ± 5.01 b	29.21 ± 6.14 c	2.49 ± 0.69 c	46.11 ± 6.78 c
	<i>B.subtilis</i>	180.14 ± 28.82 a	19.38 ± 5.16 a	45.70 ± 6.44 a	4.08 ± 0.95 a	70.83 ± 9.01 a
	<i>P.agglomerans</i>	174.20 ± 32.18 ab	19.00 ± 5.74 a	43.64 ± 7.90 a	4.00 ± 1.25 ab	66.94 ± 11.24 ab
	<i>S.indica</i>	171.35 ± 30.10 bc	17.39 ± 5.91 ab	43.24 ± 5.85 ab	3.61 ± 0.95 ab	63.89 ± 10.16 ab
	<i>F.mosseae</i>	175.00 ± 29.67 ab	19.67 ± 5.87 a	38.27 ± 5.46 b	3.16 ± 0.87 bc	59.72 ± 7.71 b
Phosphorus concentration (mg L ⁻¹)	40	207.64 ± 5.27 a	24.75 ± 2.90 a	45.92 ± 7.16 a	4.64 ± 1.12 a	70.33 ± 11.15 a
	20	168.93 ± 8.06 b	16.43 ± 2.60 b	40.25 ± 6.03 b	3.14 ± 0.80 b	60.66 ± 8.17 b
	10	143.24 ± 8.45 c	13.77 ± 1.61 c	33.87 ± 5.12 c	2.63 ± 0.61 c	53.5 ± 8.01 c

Table 2. Effects of plant growth-promoting microorganisms (PGPMs) and different phosphorus concentrations on growth parameters of hydroponically grown basil plants. Mean values marked with different letters are significantly different according Duncan's multiple range test, Values represent mean ± standard deviation of three independent replicates (n = 3).

Value sources	Df	Mean square
		Colonization (%)
Replication	2	0.38 ^{ns}
Microorganisms (a)	1	112.50 ^{ns}
Phosphorus (b)	2	1006.05**
a × b	2	568.50**
Error	10	33.52
Cv (%)		8.94

Table 3. ANOVA for the effects of arbuscular mycorrhizal fungi (AMF) and different phosphorus concentrations on root colonization percentage of hydroponically grown basil plants. ^{ns} non-significant; *significant at $p < 0.05$; **significant at $p < 0.01$.

weight, and root volume, whereas inoculation with PGPMs improved these growth parameters compared to the control. Among the treatments, *Bacillus subtilis* yielded the highest shoot and root biomass, although all microbial inoculations showed significant positive effects relative to the uninoculated control (Table 2). Arbuscular mycorrhizal fungi successfully colonized basil roots (Table 3), with colonization rates ranging from 44 to 85%, confirming the establishment of effective symbiosis under hydroponic conditions (Figs. 1 and 2).

Leaf area

The interaction between PGPMs and phosphorus concentration had a significant effect on leaf area ($P \leq 0.01$; Table 1). The largest leaf area was observed in plants inoculated with *F. mosseae* under 40 mg L⁻¹ phosphorus (Fig. 3). At this phosphorus concentration, treatments with *B. subtilis*, *P. agglomerans*, and *F. mosseae* did not differ significantly from each other but showed clear superiority compared with the control. In contrast, plants inoculated with *S. indica* exhibited no significant difference from the control. At lower phosphorus concentrations (20 and 10 mg L⁻¹), microbial treatments slightly increased leaf area compared with the control, though differences were not statistically significant.

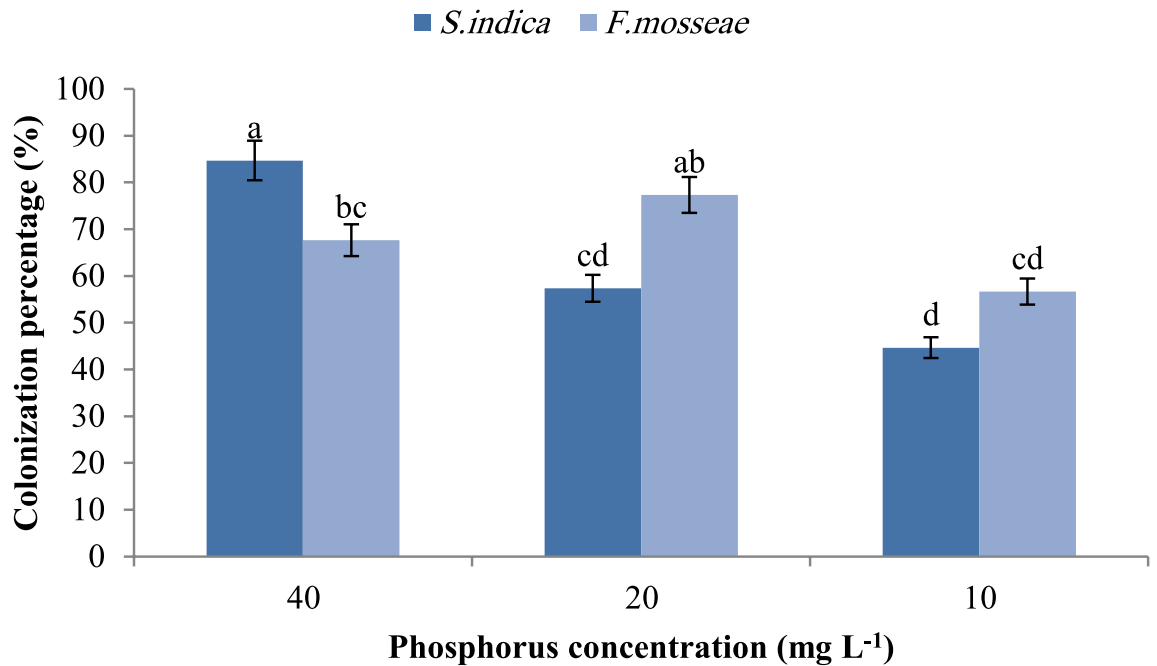


Fig. 1. Mycorrhizal root colonization (%) of hydroponically grown basil plants treated with arbuscular mycorrhizal fungi (AMF) under different phosphorus levels. Mean values marked with different letters are significantly different according to Duncan's multiple range test ($P \leq 0.01$).

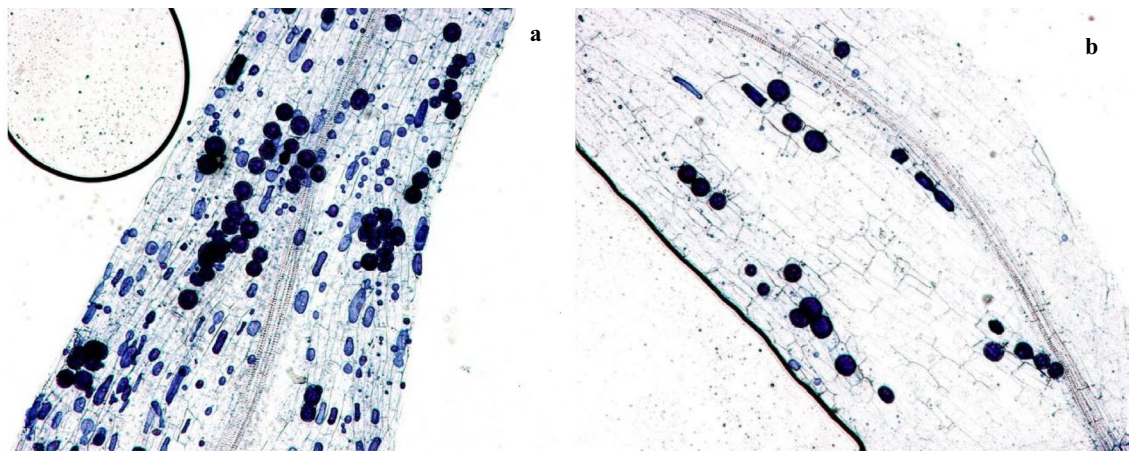


Fig. 2. Mycorrhizal root colonization of hydroponically grown basil plants inoculated with (a) *Funnelformis mosseae*, and (b) *Serendipita indica*.

Relative water content (RWC)

The interaction between PGPMs and phosphorus concentration significantly affected leaf relative water content (RWC) ($P \leq 0.01$; Table 1). Microbial inoculation enhanced RWC across all phosphorus levels, with *P. agglomerans* at 20 mg L⁻¹ P yielding the highest values, whereas control plants at 10 mg L⁻¹ P showed the lowest RWC (Fig. 4).

Phosphorus (P) concentration

The results revealed that tissue phosphorus (P) concentration in both shoots and roots of treated basil plants were significantly ($P \leq 0.01$) affected by the interaction between PGPMs and different phosphorus levels (Table 4). The highest phosphorus concentration in shoot (22.81 mg g⁻¹ DW) and root (19.64 mg g⁻¹ DW) was observed in plants inoculated with *B. subtilis* and *P. agglomerans* at 40 mg L⁻¹ P, respectively (Fig. 5 and 6). In contrast, control plants grown under 10 mg L⁻¹ P exhibited the lowest tissue P concentration in both organs. Across all treatments, a reduction in external phosphorus supply led to a marked decrease in tissue P concentration. However, inoculation with PGPMs significantly enhanced tissue P accumulation at each phosphorus level.

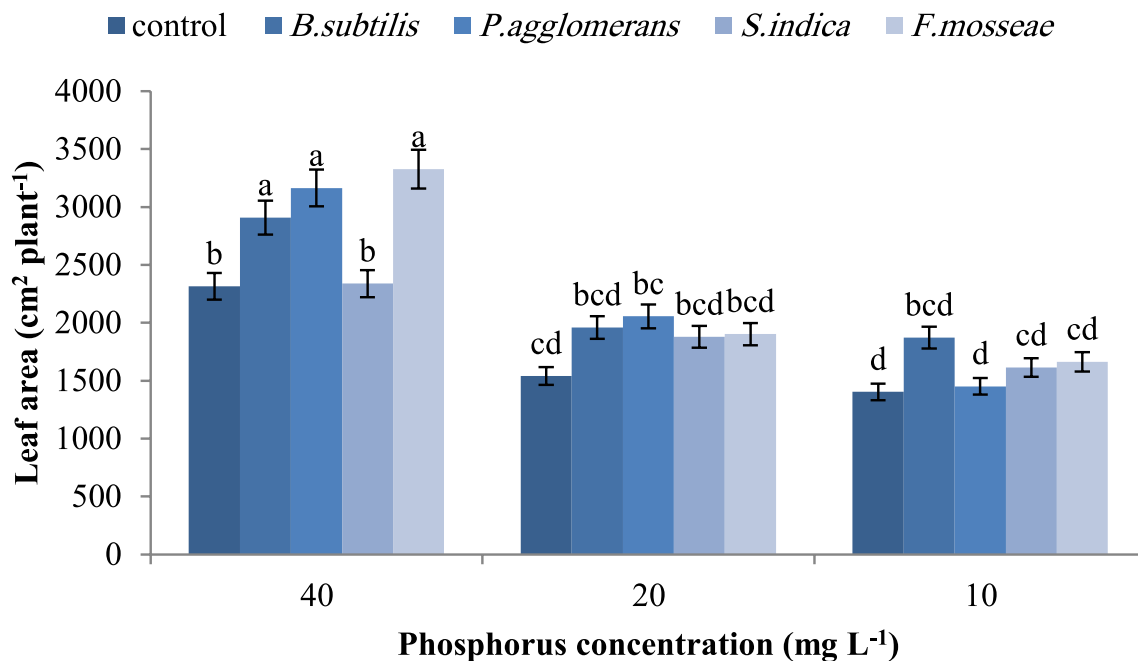


Fig. 3. Leaf area (cm² plant⁻¹) of hydroponically grown basil plants treated with PGPMs under different phosphorus levels. Mean values marked with different letters are significantly different according to Duncan's multiple range test ($P \leq 0.01$).

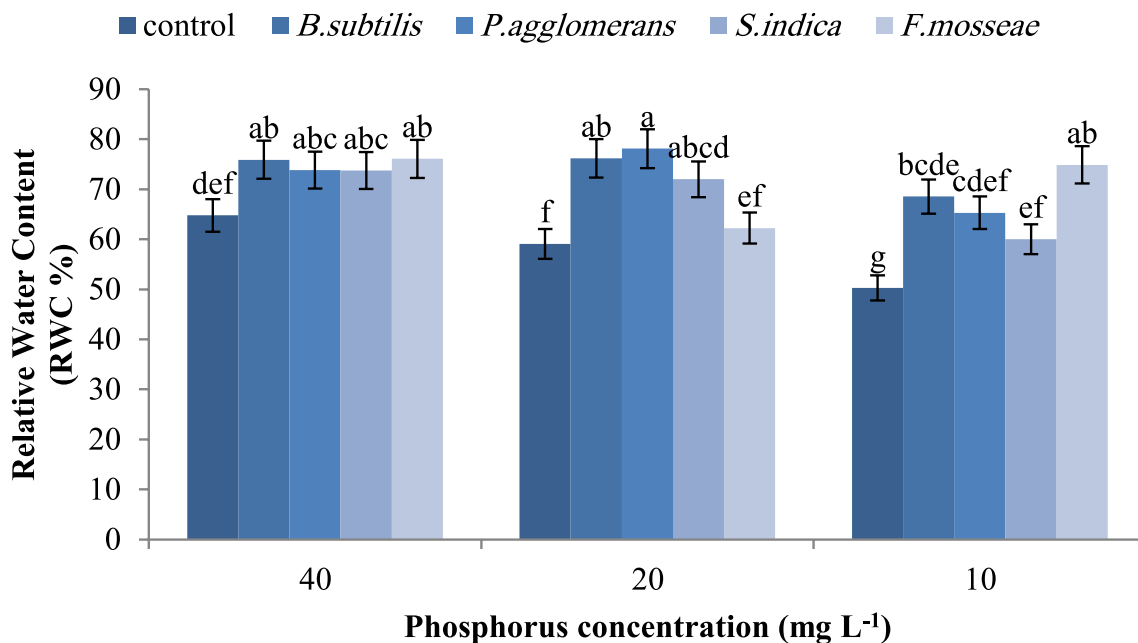


Fig. 4. Leaf relative water content of hydroponically grown basil plants treated with PGPMs under different phosphorus levels. Mean values marked with different letters are significantly different according to Duncan's multiple range test ($P \leq 0.01$).

compared with the uninoculated controls, highlighting their potential role in improving nutrient acquisition efficiency.

Potassium (K) concentration

According to the results, potassium (K) concentration in both shoots and roots of basil plants was significantly influenced ($P \leq 0.01$) by the interaction between PGPM and different phosphorus levels (Table 4). In all treatments, decreasing the external phosphorus supply was associated with an increase in tissue K concentration.

Value sources	df	Means square				
		P concentration (shoot)	P concentration (root)	K concentration (shoot)	K concentration (root)	EOs percentage (%)
Replication	2	0.008 ^{ns}	0.000 ^{ns}	1.75 ^{ns}	0.40 ^{ns}	0.0006 ^{ns}
Microorganisms (a)	4	0.145**	0.177**	5.38**	47.60**	0.0125**
Phosphorus (b)	2	2.977**	2.522**	20.82**	21.24**	0.0241**
a × b	8	0.030**	0.35**	2.18**	2.59**	0.0185**
Error	2	0.004	0.002	0.63	0.77	0.0020
Cv (%)	8	3.91	3.97	5.13	6.48	24.96

Table 4. ANOVA for the effects of PGPMs and different phosphorus levels on nutrient uptake and EOs content of hydroponically grown basil plants. ^{ns}non-significant; *significant at $p < 0.05$; **significant at $p < 0.01$.

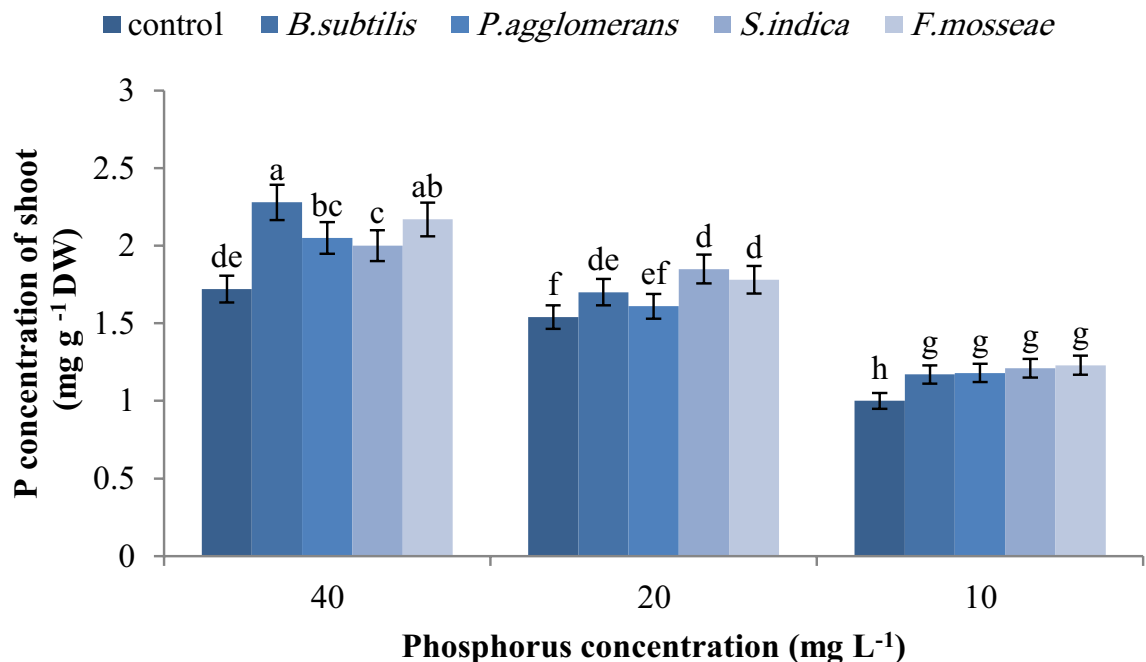


Fig. 5. Shoot phosphorus (P) concentration of hydroponically grown basil plants treated with PGPMs under different phosphorus levels, Mean values marked with different letters are significantly different according to Duncan's multiple range test ($P \leq 0.01$).

The highest K concentration in shoots ($17.0 \text{ mg g}^{-1} \text{ DW}$) was recorded in plants inoculated with *F. mosseae* under $20 \text{ mg L}^{-1} \text{ P}$, which did not differ significantly from plants inoculated with *F. mosseae* at $10 \text{ mg L}^{-1} \text{ P}$ and *P. agglomerans* at $20 \text{ mg L}^{-1} \text{ P}$ (Fig. 7). In contrast, the lowest shoot K concentration ($12.33 \text{ mg g}^{-1} \text{ DW}$) was observed in control plants grown under $40 \text{ mg L}^{-1} \text{ P}$. For root tissues, the maximum K concentration ($18.0 \text{ mg g}^{-1} \text{ DW}$) was found in plants inoculated with *S. indica* at $20 \text{ mg L}^{-1} \text{ P}$. Notably, across all phosphorus levels, root K concentration was consistently highest in plants inoculated with *S. indica* (Fig. 8).

Secondary metabolites and EOs composition

The EOs content (%) of hydroponically grown basil plants was significantly ($P \leq 0.01$) affected by the interaction between PGPMs and phosphorus (P) levels (Table 4). Inoculation with *S. indica* and *P. agglomerans* at $40 \text{ mg L}^{-1} \text{ P}$ resulted in a notable increase in EOs content, with fourfold (0.4%) and threefold (0.3%) enhancements compared with the control (0.1%), respectively (Fig. 9). Chemical profiling of the EOs indicated that estragole (methyl chavicol) was the predominant compound, followed by δ -cadinene, methyl eugenol, linalool, and β -farnesene (Table 5). Significant treatment-dependent differences in EOs composition were observed. Inoculation with *F. mosseae* at 40 and $20 \text{ mg L}^{-1} \text{ P}$ resulted in the highest concentration of Estragole (43.7% and 42.6%), respectively. Application of *B. subtilis* at $40 \text{ mg L}^{-1} \text{ P}$ resulted in the maximum δ -cadinene (16.9%), β -farnesene (7.6%), α -bergamotene (6.89%), and γ -cadinene (6.48%). Furthermore, inoculation with *P. agglomerans* at $20 \text{ mg L}^{-1} \text{ P}$ resulted in the highest observed concentrations of Methyl eugenol (10.7%). In contrast, the highest linalool concentration (8.67%) was detected in control plants grown under $10 \text{ mg L}^{-1} \text{ P}$.

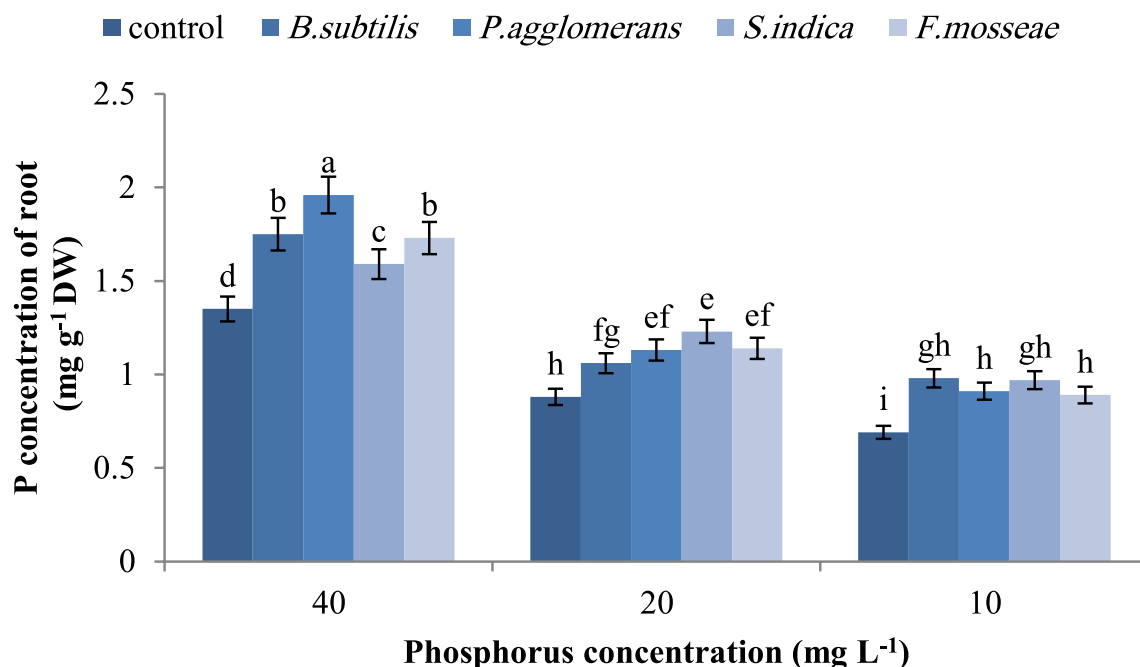


Fig. 6. Root phosphorus (P) concentration of hydroponically grown basil plants treated with PGPMs under different phosphorus levels, Mean values marked with different letters are significantly different according to Duncan's multiple range test ($P \leq 0.01$).

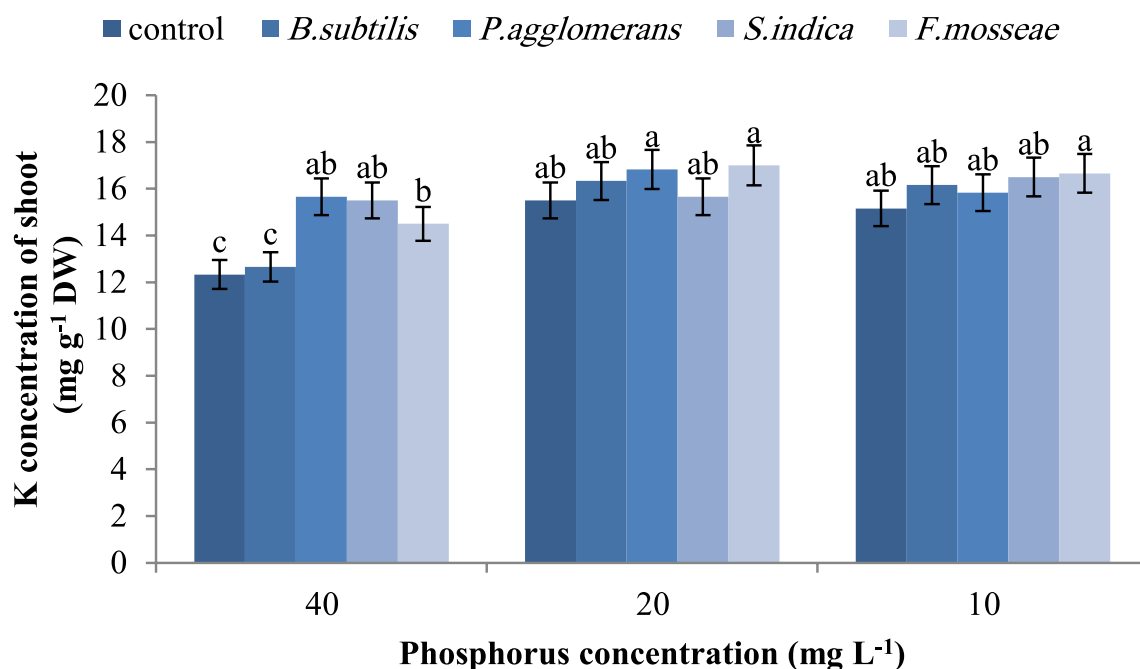


Fig. 7. Shoot potassium (K) concentration of hydroponically grown basil plants treated with PGPMs under different phosphorus levels, Mean values marked with different letters are significantly different according to Duncan's multiple range test ($P \leq 0.01$).

Discussion

The findings of this study demonstrate that beneficial microorganisms can substantially influence growth performance, nutrient dynamics, water relations, and secondary metabolite production in hydroponically grown basil. The observed levels of AMF colonization fall within the range reported for soilless culture systems and are consistent with a recent meta-analysis that identified a mean colonization rate of approximately 52%,

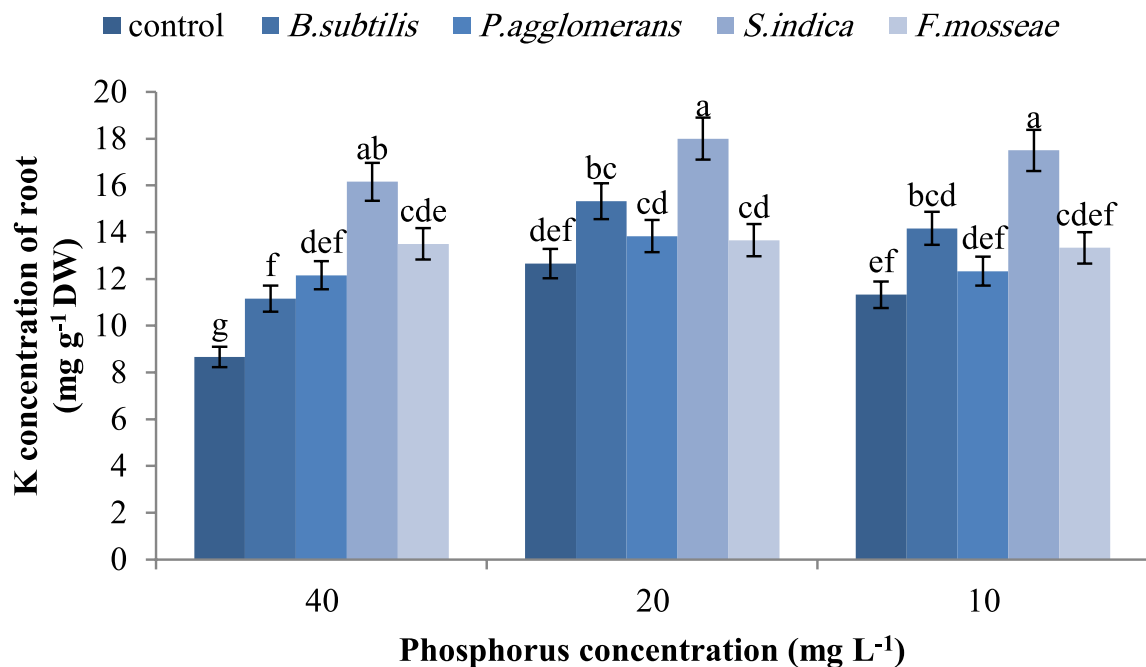


Fig. 8. Root potassium (K) concentration of hydroponically grown basil plants treated with PGPMs under different phosphorus levels, Mean values marked with different letters are significantly different according to Duncan's multiple range test ($P \leq 0.01$).

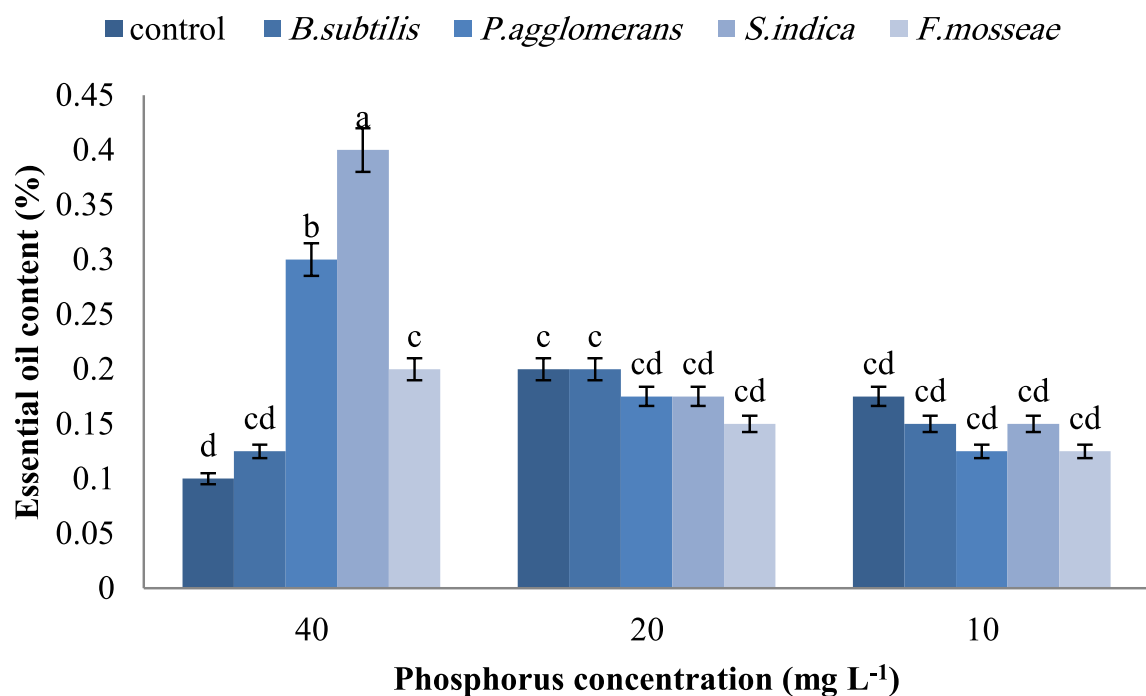


Fig. 9. EOs content (%) of hydroponically grown basil plants treated with PGPMs under different phosphorus levels, Mean values marked with different letters are significantly different according to Duncan's multiple range test ($P \leq 0.01$).

depending on inoculation procedures and substrate characteristics³¹. Such colonization efficiencies highlight the feasibility of establishing functional symbioses even in substrate-free conditions, where microbial performance has historically been considered less predictable.

Although arbuscular mycorrhizal colonization was directly verified in this study, bacterial root colonization was not quantified using culture-dependent, microscopic, or molecular approaches. Consequently, the

Treatments																
Compound	RI	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T11	T12	T13	T14	T15
1,8-Cineole	1073	–	–	2.57	–	1.01	3.7	1.89	2.01	1.61	1.89	3.88	2.0	1.9	2.54	2.0
Fenchon	1103	–	–	–	–	1.04	2.54	0.96	1.34	1.18	–	2.93	–	1.3	1.45	0.9
Linalool	1111	4.44	5.61	–	7.19	7.42	6.97	5.79	7.03	4.95	3.8	8.67	5.81	4.8	4.92	4.2
Camphor	1136	–	–	–	–	0.95	–	2.31	2.34	1.44	1.73	–	2.33	2.2	3.28	2.9
Estragole	1180	24.0	23.8	39.3	34.0	43.7	24.7	33.1	33.9	37.4	42.6	21.1	31.3	36.7	39.3	30.8
β-Citral	1202	0.99	1.38	0.81	0.95	0.82	0.71	–	0.88	–	0.65	1.2	–	–	–	–
Geraniol	1204	1.96	4.4	1.03	3.65	1.1	1.13	–	1.73	–	1.03	1.15	0.33	–	–	0.2
Geranial	1220	1.46	2.11	0.72	1.64	1.12	1.03	–	1.14	–	0.94	1.67	–	–	–	0.3
Eugenol	1271	–	–	2.31	–	–	–	0.87	1.25	1.79	–	–	–	0.4	–	–
Methyl Eugenol	1303	2.44	1.92	3.16	7.87	3.35	5.3	4.41	10.7	5.92	2.66	5.99	6.11	7.3	4.95	3.9
Bornyl acetate	1328	–	–	–	–	0.88	0.77	0.88	1.0	0.62	0.71	–	0.67	0.7	0.77	0.4
β-Damascenone	1401	–	–	0.83	–	0.51	0.77	–	–	0.64	–	–	–	–	–	–
γ-Elementene	1477	1.17	–	1.12	0.78	0.88	1.12	1.26	1.03	1.14	0.67	1.14	1.33	1.0	0.81	0.8
Copaene	1498	–	–	0.04	–	0.27	–	0.38	0.31	0.27	–	–	–	–	–	0.4
β-Elementene	1507	4.4	4.03	3.29	1.77	1.95	2.27	4.48	2.81	1.86	1.86	3.3	3.54	3.2	2.75	2.8
β-Caryophyllene	1522	3.54	5.51	3.43	2.88	2.65	3.41	3.76	2.83	3.06	3.02	3.43	3.66	2.9	2.22	2.6
β-Cubebene	1527	1.0	1.08	0.97	–	0.69	0.84	0.89	0.74	0.75	0.56	0.71	0.99	0.7	0.55	0.5
α-Bergamotene	1531	3.38	6.89	5.25	6.31	3.75	5.88	5.43	2.99	4.18	4.81	5.54	3.77	5.1	4.39	5.1
α-Guaiene	1532	0.93	–	–	–	0.6	0.65	–	0.64	0.49	0.59	–	0.83	0.8	0.58	–
β-Farnesene	1540	3.61	7.6	3.52	4.87	3.22	3.48	3.51	2.75	3.29	3.74	2.77	4.12	2.0	2.74	3.2
cis-Muurolo	1545	1.76	1.29	1.39	0.92	0.85	0.97	1.34	0.95	0.93	0.87	0.94	1.07	0.8	0.87	0.6
Germacrene D	1555	2.04	2.19	1.73	–	0.71	–	1.82	1.29	1.73	–	–	1.65	–	1.57	–
α-Bulnesene	1566	2.0	1.78	1.75	1.22	1.08	1.43	1.59	1.17	0.8	1.03	1.34	1.52	1.6	1.2	1.2
γ-Cadinene	1571	4.58	6.48	3.67	4.86	3.26	3.37	3.54	2.19	3.49	4.0	3.58	3.72	3.6	2.88	3.6
Calamenene	1574	1.55	1.27	–	1.0	0.98	0.96	1.06	0.8	1.18	1.22	0.99	1.23	1.1	0.96	1.3
Spathulenol	1602	–	2.7	1.88	2.2	1.63	2.17	2.05	1.51	1.85	2.48	2.81	2.12	2.3	2.34	4.0
Caryophyllene oxide	1605	1.22	–	1.14	–	0.87	1.28	1.16	0.7	1.07	1.81	1.95	1.04	1.1	1.26	1.4
Humulene epoxide	1616	–	–	–	–	–	0.77	0.65	–	0.55	0.8	–	0.71	0.8	0.84	1.1
α-Cubebene	1619	2.94	2.66	1.92	2.22	1.76	1.97	1.83	1.36	1.98	2.08	2.25	2.27	1.8	1.79	2.2
δ-Cadinene	1633	11.7	16.9	7.98	14.01	7.89	8.89	7.69	5.97	8.24	8.6	9.78	9.64	7.7	7.82	9.6
α-Cadinol	1638	1.59	–	1.71	0.95	1.24	–	1.42	–	1.28	–	1.3	1.65	–	–	1.9
β-Eudesmol	1649	–	–	–	–	–	1.88	–	1.2	–	1.09	–	–	1.0	1.55	–
Palmitic acid	1827	4.61	–	1.2	–	0.52	2.27	1.67	0.7	1.75	1.0	1.87	1.47	1.0	0.79	1.6
hexahydrofarnesyl acetone	1883	1.16	–	0.53	–	0.41	0.58	0.61	0.44	0.52	0.56	0.78	0.67	0.5	0.52	0.5
Phytol	2115	0.96	–	0.5	–	–	0.85	0.64	0.44	0.6	–	–	0.55	0.4	0.42	0.6
Neophytadiene	2163	0.67	–	–	–	0.32	0.46	0.38	–	0.35	–	–	–	–	–	0.3

Table 5. Relative abundance of major compounds in the EOs of aerial parts of *Ocimum basilicum* L. as affected by PGPMs and different phosphorus concentrations. RI: Retention Index; T1: control, P40 (mg L⁻¹) T2: *B. subtilis*, P40 T3: *P. agglomerans*, P40 T4: *S. indica*, P40 T5: *F. mosseae*, P40 T6: control, P20 (mg L⁻¹) T7: *B. subtilis*, P20 T8: *P. agglomerans*, P20 T9: *S. indica*, P20 T10: *F. mosseae*, P20 T11: control, P10 (mg L⁻¹) T12: *B. subtilis*, P10 T13: *P. agglomerans*, P10 T14: *S. indica*, P10 T15: *F. mosseae*, P10.

effectiveness of bacterial inoculation was inferred from consistent and treatment-specific phenotypic and physiological responses, including enhanced biomass accumulation, improved nutrient uptake, and altered metabolic profiles in PGPM-treated plants compared with non-inoculated controls across phosphorus levels. Such responses are widely accepted as indirect indicators of successful root interaction in PGPM studies, particularly under controlled cultivation systems. Nevertheless, direct confirmation of bacterial colonization remains an important limitation of the present work. Future studies integrating plate counting, fluorescence-based microscopy, or molecular techniques such as qPCR will be valuable for verifying colonization dynamics and strengthening mechanistic interpretation of plant–bacteria interactions in hydroponic systems.

The positive effects of microbial inoculation on biomass accumulation and physiological responses align with extensive literature demonstrating that PGPMs enhance plant growth through multiple direct and indirect mechanisms. These include modulation of hormonal signaling, biological nitrogen fixation, phosphate solubilization, siderophore production, and the release of volatile organic compounds^{1,13,32}. *Bacillus* species, in particular, are widely recognized for producing phytohormones, siderophores, and lipopeptides that simultaneously promote growth and suppress phytopathogens^{33–36}. Parallel findings have been reported for other crop species cultivated hydroponically or under nutrient-limited conditions, where inoculation

with *Azotobacter*, *Azospirillum*, *Bacillus*, and AMF consistently enhanced biomass production^{37–40}. Previous studies in basil similarly demonstrated the growth-promoting effects of microbial inoculants, supporting the generalizability of our results^{41,42}.

The improvements in leaf area following microbial inoculation further corroborate earlier evidence from onion, lettuce, banana, and tomato, where PGPMs enhanced leaf expansion through increased nutrient acquisition and hormonal regulation^{37,43–46}. Physiologically, such enhancements may reflect improved water and nutrient status as well as increased synthesis of IAA and other growth-regulating metabolites.

Changes in relative water content (RWC) suggest that PGPMs improve water relations through both physiological and molecular processes. Enhanced stomatal performance, improved leaf water potential, and increased osmolyte accumulation have been reported as primary contributors to improved water status under microbial treatments^{47,48}. Molecular studies indicate that PGPMs can upregulate aquaporin gene families (PIPs, TIPs), thereby increasing root hydraulic conductivity⁴⁹. AMF hyphae also access water from pores too small for root hairs, contributing to improved hydration⁵⁰. Reports of increased RWC in mycorrhizal soybean and other crops^{51,52}. Further support these mechanisms and align with the responses observed in basil.

Enhanced phosphorus acquisition in inoculated plants reflects the combined contributions of phytase activity, organic acid production, root architectural modifications, and hyphal foraging by AMF. *P. agglomerans* and related species are well known for phytase synthesis, enabling the mobilization of phytate—the dominant form of organic P in many substrates⁵³. AMF-mediated increases in alkaline and acid phosphatase activities and improvements in N, P, and K uptake have been documented in several plant species⁵⁴. Hyphal networks play a particularly important role in expanding the effective absorptive surface area of roots and accessing P bound within microsites unavailable to root hairs^{55,56}. Organic acid-mediated chelation, siderophore production, and HCN generation further contribute to phosphorus solubilization and metal sequestration, enhancing nutrient availability in the rhizosphere^{14,36,48}. Together, these findings suggest that microbial inoculants employ a complementary suite of biochemical and morphological strategies to improve P-use efficiency in hydroponic basil.

Potassium uptake was also enhanced by microbial inoculation, consistent with the activity of potassium-solubilizing bacteria that mobilize K from insoluble minerals through acidification, chelation, and exopolysaccharide-mediated mineral dissolution⁵⁷. Several strains of *Enterobacter* and *Pantoea* have demonstrated strong K-solubilizing capabilities⁵⁸. AMF further contribute indirectly by modifying root morphology and altering exudate profiles, thereby improving K absorption^{59,60}. These findings underscore the importance of microbial–nutrient interactions in overcoming the limitations associated with low solubility of K-bearing fertilizers.

Microbial inoculation also exerted pronounced effects on EOs content and secondary metabolite profiles. In this study, EOs were extracted from dried plant material, which is a widely accepted approach in basil and other aromatic plant research. Although some highly volatile constituents may be partially lost during air-drying, drying at room temperature in the shade ensures minimal thermal degradation and provides a uniform physiological baseline for all samples, particularly under controlled hydroponic cultivation. Expressing EOs yield on a dry-weight basis allows for more reliable and reproducible comparisons across treatments, as fresh weight would be strongly influenced by tissue water content. Any potential volatilization would have occurred uniformly across all samples and therefore does not affect relative treatment comparisons. Numerous studies have shown that PGPRs and AMF can enhance the synthesis of alkaloids, terpenoids, phenolics, and other bioactive compounds in medicinal plants^{26,61–63}. The enhancement of terpenoid constituents such as estragole, linalool, and methyl eugenol observed in basil may be attributed to induced systemic resistance (ISR) activation, improved nutrient status, and morphological modifications such as increased glandular trichome density^{64,65}. AMF-induced increases in trichome density and associated metabolite production have been reported for *Artemisia*, *Origanum*, and *Ocimum* species⁶⁶. The observed influence of phosphorus nutrition on secondary metabolite pathways is also consistent with earlier findings indicating that P availability governs isoprenoid biosynthesis through its roles in ATP and NADPH generation and acetyl-CoA metabolism¹⁹. Collectively, these observations suggest that the combined effects of improved nutrient acquisition, morphological adaptations, and biochemical signaling drive the enhanced EOs yield and altered composition associated with PGPM inoculation.

From an applied perspective, the use of microbial inoculants offers a promising strategy to enhance yield and quality in medicinal plant production while reducing reliance on chemical inputs. The diverse mechanisms employed by beneficial microorganisms—including nutrient solubilization, hormone production, ISR induction, and stress mitigation—align with global efforts toward sustainable, environmentally responsible agriculture⁶⁷.

This study demonstrates the potential of PGPM inoculation combined with optimized phosphorus management to enhance basil growth, nutrient uptake, and essential oil production under controlled hydroponic conditions. The methodologies developed here could be adapted to larger hydroponic systems, offering opportunities for commercial or semi-commercial production. Scaling up would require adjustments such as maintaining uniform distribution of nutrients and microbial inoculants, ensuring consistent root colonization, and monitoring microbial population dynamics, biofilm formation, and interactions with system components. Continuous aeration, improved mixing, automated nutrient delivery, and controlled-release formulations of PGPMs may be necessary to sustain microbial activity over longer production cycles. Additionally, tighter control of environmental parameters, including temperature, oxygen availability, and solution pH, would be essential to preserve both microbial functionality and plant performance.

Future studies should focus on pilot-scale applications and long-term monitoring to validate these approaches under commercial conditions. By addressing these challenges, the findings from this research can inform the development of efficient, scalable, and sustainable hydroponic production systems for basil and other medicinal or aromatic plants, bridging the gap between controlled experiments and practical agricultural applications.

Conclusion

The findings of this study demonstrate that no single PGPM exhibited universal superiority across all evaluated traits; rather, their beneficial effects were trait-specific. *B. subtilis* was the most effective in enhancing biomass production and shoot phosphorus accumulation, while *F. mosseae* maximized leaf area and estragole content in EOs. *P. agglomerans* contributed markedly to improved relative water content and root phosphorus concentration, whereas *S. indica* proved most efficient in increasing root potassium uptake and boosting EOs percentage. These results highlight the functional diversity of PGPMs and underscore the importance of selecting specific microbial inoculants tailored to targeted agronomic or phytochemical objectives. Collectively, the study confirms that integrating PGPMs into hydroponic basil cultivation represents a sustainable biotechnological strategy to enhance growth performance, optimize nutrient acquisition, and improve the yield and quality of secondary metabolites, offering a promising eco-friendly alternative to chemical inputs in medicinal plant production. Future research should focus on field-level validation and the exploration of specific microbial strains to maximize their potential as biotechnological tools in medicinal plant production.

Materials and methods

Experimental site and experimental design

The experiment was carried out in the hydroponic research greenhouse of the Department of Horticultural Science, University of Tabriz, Iran (38°27' N, 46°27' E; 1576 m a.s.l.). A factorial experiment was arranged in a completely randomized design (CRD) with three replications. Treatments included three phosphorus levels in Hoagland nutrient solution (10, 20, and 40 mg L⁻¹), representing deficient, moderate, and adequate concentrations commonly used in hydroponic plant cultivation, and five microbial inoculation treatments: *Funneliformis mosseae*, *Serendipita indica*, *Bacillus subtilis*, *Pantoea agglomerans*, and non-inoculated control. The non-inoculated control was included to isolate the effects of microbial inoculation under identical nutrient and substrate conditions. In the non-inoculated control, plants received the same substrate and nutrient solutions as inoculated plants, but the substrate was autoclaved to exclude microbial activity. This factorial design allowed the specific effects of microbial inoculation at each phosphorus level to be assessed, ensuring that observed differences could be attributed solely to the microbial treatments.

The experiment was conducted under controlled greenhouse conditions, and treatments were randomly assigned to minimize potential confounding variables. No issues with replication or confounding factors were observed, and environmental conditions, nutrient solutions, and substrate composition were maintained uniformly across all treatments.

Microbial inoculants

All microbial inoculants were obtained from the Soil Biology Laboratory, University of Tabriz. For the bacterial treatments: *Bacillus subtilis* and *Pantoea agglomerans* were cultured separately in nutrient broth (NB) at 28 °C for 24 h under constant shaking at 130 rpm until reaching the logarithmic growth phase, which was monitored via optical density (OD₆₀₀). The bacterial suspensions were centrifuged, and the resulting pellets were resuspended in sterile distilled water to a final concentration of 10⁸ CFU mL⁻¹ for each bacterium, as determined by serial dilution and plate count methods.

Inoculum of *Funneliformis mosseae* was produced using a trap culture system with sorghum (*Sorghum bicolor*) as the host plant. A phosphorus-deficient sandy loam soil was used to produce the arbuscular mycorrhizal fungal (AMF) inoculant. The soil had the following characteristics: organic C, 0.17% (w/w); available K, 198 mg kg⁻¹; available P (Olsen), 3 mg kg⁻¹; electrical conductivity, 1.9 dS m⁻¹; and pH, 7.56. Prior to use, the soil was sterilized (121 °C, 1 bar, 1 h), and then used in as a carrier/substrate for the AMF inoculant. The inoculum contained spores, hyphae, and colonized root fragments, and was standardized to 10⁷–10⁸ infective propagules per gram of substrate.

The *Serendipita indica* inoculum was produced by initially culturing the fungus on potato dextrose agar (PDA) at 28 °C under sterile conditions. After a 10-day incubation period, the fungal colony surface (containing both mycelia and spores) was carefully scraped using a sterile scalpel. The resulting biomass was then thoroughly mixed into the sterilized soil substrate to create the final solid inoculant. To generate a solid-substrate inoculum in a form comparable to the AMF inoculant, the fungal mass was thoroughly mixed into the same sterilized sandy loam soil described above. The final fungal inoculums were standardized based on propagule density, achieving a target concentration of 10⁷ to 10⁸ propagules per gram of carrier. All inoculants were freshly prepared prior to application to ensure viability and consistency across treatments.

Plant material and growth conditions

Seeds of basil (*Ocimum basilicum* L., Tabriz landrace) were obtained from a certified commercial agricultural supplier (Golbargapamchal.co, Tabriz, Iran). A voucher specimen of *O. basilicum* L. was prepared and deposited in the Herbarium of the Faculty of Agriculture, University of Tabriz, under voucher number 16732, and plant identification was carried out by specialists of this herbarium.

Seeds were sown in trays containing a sterile peat–perlite mixture (1:1, v/v). The substrate was autoclaved prior to sowing to eliminate native microorganisms and ensure that root colonization resulted exclusively from the applied microbial inoculants. Seedlings were irrigated uniformly with half-strength Hoagland nutrient solution (pH 6.2) during the establishment period.

After establishment, seedlings were transplanted into 2-L pots filled with a cocopeat–perlite mixture (1:3, v/v) and maintained under controlled greenhouse conditions. Air temperature was maintained at 22 ± 2 °C during the day and 18 ± 2 °C at night, with relative humidity of 60 ± 5% and a 16 h photoperiod under natural sunlight.

Environmental parameters were controlled using automated ventilation and shading systems to minimize temperature and humidity fluctuations.

Inoculation procedure and phosphorus treatments

Microbial inoculants were applied individually. Fungal inoculants were incorporated into the peat–perlite growth substrate at sowing at a rate of 10 g inoculum per tray cell, ensuring uniform distribution within the substrate. The fungal inoculum had a propagule density of 10^7 – 10^8 per gram, providing sufficient potential for effective root colonization. Bacterial suspensions (1 mL per tray cell) were applied directly to the root zone seven days after germination. Non-inoculated control plants received the same amount of autoclaved fungal substrate and sterile distilled water to maintain equivalent physical conditions while excluding microbial activity, thereby isolating the effects of microbial inoculation under identical nutrient and environmental conditions.

Phosphorus treatments were initiated immediately after transplanting. From this stage onward, plants were irrigated exclusively with modified Hoagland nutrient solutions containing the designated phosphorus concentrations (10, 20, or 40 mg P L⁻¹), corresponding to deficient, moderate, and adequate phosphorus levels. Phosphorus concentrations were maintained constant throughout the experimental period until harvest. Nutrient solutions were freshly prepared every three days, and electrical conductivity (EC) and pH were monitored daily and adjusted to 6.2 to ensure stable nutrient availability.

Verification of root colonization

Root colonization by the inoculated fungi, *F. mosseae* and *S. indica*, was verified following the clearing and staining procedures described by^{68,69}. Root samples were cleared in KOH, stained with trypan blue, and examined microscopically. Percentage colonization was quantified using the gridline intersect method, allowing objective and reproducible quantification. Root colonization by bacterial inoculants (*B. subtilis* and *P. agglomerans*) was inferred based on observed phenotypic and physiological responses, including enhanced growth and nutrient uptake, compared to non-inoculated controls.

Growth measurements and sampling

12 weeks after sowing, at full flowering stage, basil plants were harvested. Harvesting at full bloom is considered optimal, as higher essential oil yield has been reported at this developmental stage compared with earlier growth stages^{70,71}. Shoots and roots were separated and weighed for fresh biomass using a digital precision balance (± 0.01 g).

Root volume was determined immediately after harvest and prior to any drying procedures using the water displacement method. Clean root systems were gently blotted to remove excess surface moisture and submerged in a graduated cylinder containing a known volume of water, and the increase in water level was recorded as root volume (mL). All measurements were performed using three biological replicates per treatment.

Dry weight was determined after air-drying samples at room temperature in the shade to constant weight. This drying method was employed to minimize thermal degradation and volatilization of EOs constituents and is commonly used in EOs studies of basil and other aromatic plants^{72–75}. Any potential loss of highly volatile constituents during drying would have occurred uniformly across all samples and therefore does not affect the relative comparison among treatments.

Determination of leaf area

Leaf area was measured using a leaf area meter (LI-3100, LI-COR Biosciences, USA). Fully expanded leaves were detached and immediately scanned to minimize dehydration and shrinkage. Leaves were gently flattened and arranged without overlap on the scanner conveyor to ensure accurate detection of leaf margins. The instrument was calibrated according to the manufacturer's instructions prior to measurements. Leaf area was recorded for each plant and expressed as cm² plant⁻¹. All measurements were performed in triplicate to reduce experimental variability and potential measurement error.

Determination of relative water content (RWC)

Leaf relative water content (RWC) was determined following the method described by⁷⁶. Fresh weight (FW) was recorded immediately after leaf sampling to minimize water loss due to transpiration. Leaves were rehydrated in distilled water at 4 °C for 24 h to obtain turgid weight (TW). Prior to weighing, leaf surfaces were gently blotted with filter paper to remove excess surface moisture and reduce weighing errors. Subsequently, samples were oven-dried at 60 °C for 24 h until constant weight to determine dry weight (DW). All measurements were conducted in triplicate to reduce experimental variability. Relative water content was calculated using the equation:

$$RWC(\%) = \frac{(FW - DW)}{(TW - DW)} \times 100$$

Determination of phosphorus concentration

Plant shoot and root samples were thoroughly washed with distilled water to remove surface contaminants, oven-dried at 70 °C to constant weight, ground to a fine powder, and subjected to wet acid digestion following⁷⁷. Digestion was performed using analytical-grade reagents. To minimize contamination, all glassware and plasticware were soaked in 10% (v/v) nitric acid for at least 24 h and thoroughly rinsed with distilled water prior to use. The digested solutions were filtered, brought to a known volume with distilled water and used for both phosphorus and potassium determinations.

Phosphorus concentration in the digested extracts was determined colorimetrically using the phosphovanado-molybdate method. In this assay, orthophosphate ions react with vanadate–molybdate reagent in an acidic medium to form a yellow phosphovanado-molybdate complex with maximum absorbance at 430 nm. For analysis, 0.5 mL of digested extract was mixed with 0.5 mL of nitrovanado–molybdate reagent and 2 mL of distilled water. After incubation for 1 h at room temperature, absorbance was measured at 430 nm using a UV–visible spectrophotometer (UV-2100, Unico, UK). Quantification was performed using an external calibration curve prepared from potassium dihydrogen phosphate (KH_2PO_4) standard solutions with concentrations of 0, 2, 4, 6, 8, and 10 mg L^{-1} P. Phosphorus concentrations in the samples were calculated using the linear regression equation derived from the standard curve.

Determination of potassium concentration

Potassium concentration in plant shoot and root samples was determined using a flame photometric method following the general procedure described by⁷⁸. After acid digestion, potassium was quantified using a flame photometer (Corning 410, UK) equipped with a propane-air flame system. Prior to measurement, a 0.5 mL aliquot of the digested plant extract was diluted with 4.5 mL of distilled water to ensure that the emission intensity fell within the linear detection range of the instrument.

Potassium atoms were excited in the flame and emitted characteristic radiation at 766.5 nm, and the emission intensity was recorded by the photometer. Instrument calibration was carried out using potassium standard solutions prepared from potassium chloride (KCl). A series of standards (0, 2, 4, 6, 8, and 10 mg L^{-1} K) was prepared to generate a calibration curve and potassium concentrations in the samples were calculated accordingly. Calibration standards were freshly prepared for each analytical run, and all measurements were conducted in triplicate.

Extraction and analysis of the essential oils

The EOs content of aerial parts of treated basil plants at the full flowering stage was determined. EOs were extracted from 40 g of dried plant material by hydrodistillation using a Clevenger-type apparatus for 3 h. EOs content was calculated on a dry weight basis and expressed as percentage (v/w). The oils were dried over anhydrous sodium sulfate, stored in dark glass vials, and maintained at 4 °C until analysis⁷⁹. EOs extraction was performed using three independent biological replicates per treatment and extracted oils were subsequently pooled into a composite sample for each treatment prior to chemical analysis, as commonly practiced in essential oil research^{80,81}. EOs content (% v/w) was calculated as follows:

$$\text{EOs content}(\%v/w) = \frac{\text{volume of extracted oil (mL)}}{\text{dry weight of plant material (g)}} \times 100$$

The chemical composition of the EOs was analyzed using an Agilent 7890B gas chromatograph coupled with an Agilent 5977A mass spectrometer (GC–MS), equipped with an HP-5MS UI capillary column (30 m × 0.25 mm i.d., 0.25 μm film thickness). Helium (99.99% purity) was used as the carrier gas at a constant flow rate of 1 mL min^{-1} (linear velocity 36.4 cm s^{-1}). The column temperature was programmed from 60 to 325 °C, with the injection port maintained at 280 °C. One microliter of EOs was injected per run. Constituents were identified based on retention indices, mass spectral data, and comparison with NIST and Wiley libraries.

Statistical analysis

Data were analyzed using analysis of variance (ANOVA) appropriate for a factorial experiment in a completely randomized design, with three replications. Prior to analysis, data were examined for potential outliers using boxplot analysis based on the $1.5 \times$ interquartile range (IQR) criterion, and no statistical outliers were detected. The assumptions of the ANOVA model, including normality and homogeneity of variance, were verified and met. Mean comparisons were performed using Duncan's multiple range test at $P \leq 0.05$. All replications were conducted under uniform experimental conditions, and no confounding variables were identified. Statistical analyses were conducted using SPSS version 16.0. Standard deviations are reported in tables and graphs using error bars.

Data availability

The datasets generated or analyzed during the current study are available from the corresponding author on reasonable request.

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Author contributions

F. M. performed the experiments, collected and analyzed the data, and drafted the manuscript. S. B. conceived and designed the study and assisted with data analysis. S. A. S. contributed to methodology and manuscript editing. M. S. prepared microbial inocula and revised the manuscript. All authors approved the final version and are accountable for the work.

Competing interests

The authors declare no competing interests.

Additional information

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